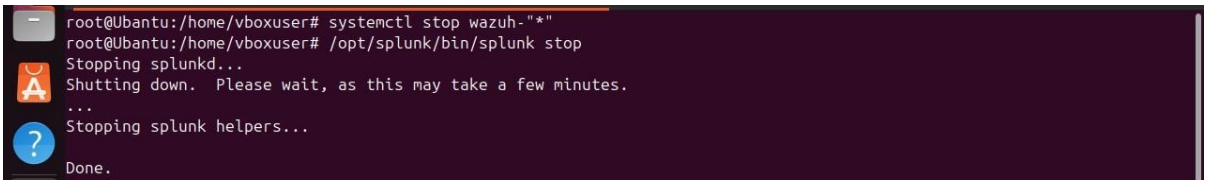
Zeek logs on ELK

Zeek (formerly called Bro) is a free tool used to monitor network security. It helps track network activity and detect any security issues by analyzing traffic in real-time.

The ELK Stack is a set of three tools—Elasticsearch, Logstash, and Kibana—that work together to manage and analyze log data. It collects logs from different sources, processes them, and displays them in a way that makes it easy to understand and spot important patterns or issues in real-time.

systemctl stop wazuh-“*”

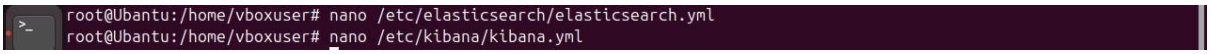
/opt/splunk/bin/splunk stop



- The first command stops all Wazuh services, which are used for security monitoring and incident response.
- The second command stops the Splunk service, which is used for log management and data analysis.

nano /etc/elasticsearch/elasticsearch.yml

nano /etc/kibana/kibana.yml



Both commands are used to change the settings for Elasticsearch and Kibana. You can adjust their configurations to fit your needs. After making changes, you usually save the files and restart the services to apply the changes.

Elasticsearch.yml file changes

```
root@Ubantuu: /home/vboxuser
  GNU nano 7.2                    /etc/elasticsearch/elasticsearch.yml
# Make sure that the heap size is set to about half the memory available
# on the system and that the owner of the process is allowed to use this
# limit.
#
# Elasticsearch performs poorly when the system is swapping the memory.
#
# ---------------------------------- Network -----------------------------------
#
# By default Elasticsearch is only accessible on localhost. Set a different
# address here to expose this node on the network:
#
network.host: 192.168.1.65
#
# By default Elasticsearch listens for HTTP traffic on the first free port it
# finds starting at 9200. Set a specific HTTP port here:
#
http.port: 9200
#
# For more information, consult the network module documentation.
#
# --------------------------------- Discovery ----------------------------------
#
# Pass an initial list of hosts to perform discovery when this node is started:

^G Help      ^O Write Out  ^W Where Is   ^K Cut       ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify    ^/ Go To Line M-E Redo
```

Kibana.yml file changes

```
root@Ubantuu: /home/vboxuser
  GNU nano 7.2                    /etc/kibana/kibana.yml
# Kibana is served by a back end server. This setting specifies the port to use.
server.port: 5601

# Specifies the address to which the Kibana server will bind. IP addresses and host names are both >
# The default is 'localhost', which usually means remote machines will not be able to connect.
# To allow connections from remote users, set this parameter to a non-loopback address.
server.host: "192.168.1.65"

# Enables you to specify a path to mount Kibana at if you are running behind a proxy.
# Use the `server.rewriteBasePath` setting to tell Kibana if it should remove the basePath
# from requests it receives, and to prevent a deprecation warning at startup.
# This setting cannot end in a slash.
#server.basePath: ""

# Specifies whether Kibana should rewrite requests that are prefixed with
# `server.basePath` or require that they are rewritten by your reverse proxy.
# This setting was effectively always `false` before Kibana 6.3 and will
# default to `true` starting in Kibana 7.0.
#server.rewriteBasePath: false

# Specifies the public URL at which Kibana is available for end users. If
# `server.basePath` is configured this URL should end with the same basePath.
#server.publicBaseUrl: ""
                                  [ Read 115 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut       ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify    ^/ Go To Line M-E Redo
```

```
  GNU nano 7.2                    /etc/kibana/kibana.yml *
# This setting cannot end in a slash.
#server.basePath: ""

# Specifies whether Kibana should rewrite requests that are prefixed with
# `server.basePath` or require that they are rewritten by your reverse proxy.
# This setting was effectively always `false` before Kibana 6.3 and will
# default to `true` starting in Kibana 7.0.
#server.rewriteBasePath: false

# Specifies the public URL at which Kibana is available for end users. If
# `server.basePath` is configured this URL should end with the same basePath.
#server.publicBaseUrl: ""

# The maximum payload size in bytes for incoming server requests.
#server.maxPayload: 1048576

# The Kibana server's name.  This is used for display purposes.
#server.name: "your-hostname"

# The URLs of the Elasticsearch instances to use for all your queries.
elasticsearch.hosts: ["http://192.168.1.65:9200"]

# Kibana uses an index in Elasticsearch to store saved searches, visualizations and

^G Help      ^O Write Out  ^W Where Is   ^K Cut       ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify    ^/ Go To Line M-E Redo
```

zeekctl check

zeekctl deploy

```
root@workstation:/home/vboxuser# zeekctl check
zeek scripts are ok.
root@workstation:/home/vboxuser# zeekctl deploy
checking configurations ...
installing ...
removing old policies in /opt/zeek/spool/installed-scripts-do-not-touch/site ...
removing old policies in /opt/zeek/spool/installed-scripts-do-not-touch/auto ...
creating policy directories ...
installing site policies ...
generating standalone-layout.zeek ...
generating local-networks.zeek ...
generating zeekctl-config.zeek ...
generating zeekctl-config.sh ...
stopping ...
stopping zeek ...
creating crash report for previously crashed nodes: zeek
starting ...
starting zeek ...
root@workstation:/home/vboxuser#
```

The zeekctl check command is a preliminary step to ensure that your configurations are correct, while the zeekctl deploy command applies those configurations and starts the Zeek instance, making it operational with the updated settings.

systemctl restart elasticsearch

systemctl restart logstash systemctl

restart kibana

```
root@Ubantu:/home/vboxuser# systemctl restart elasticsearch
root@Ubantu:/home/vboxuser# systemctl restart logstash
root@Ubantu:/home/vboxuser# systemctl restart kibana
root@Ubantu:/home/vboxuser#
```

The three commands (systemctl restart elasticsearch, systemctl restart logstash, and systemctl restart kibana) are used to restart each part of the ELK Stack.
Restarting these services is important to apply changes, fix problems, or after updates.
Using systemctl makes sure the services are properly managed by the system, so you can control them easily.

nano /etc/filebeat/filebeat.yml

```
GNU nano 7.2                    /etc/filebeat/filebeat.yml
filebeat.inputs:

# Each - is an input. Most options can be set at the input level, so
# you can use different inputs for various configurations.
# Below are the input specific configurations.

# filestream is an input for collecting log messages from files.
- type: filestream

  # Unique ID among all inputs, an ID is required.
  id: my-filestream-id

  # Change to true to enable this input configuration.
  enabled: false

  # Paths that should be crawled and fetched. Glob based paths.
  paths:
    - /opt/zeek/share/zeek/site/local.zeek/*.log
    #- c:\programdata\elasticsearch\logs\*
```

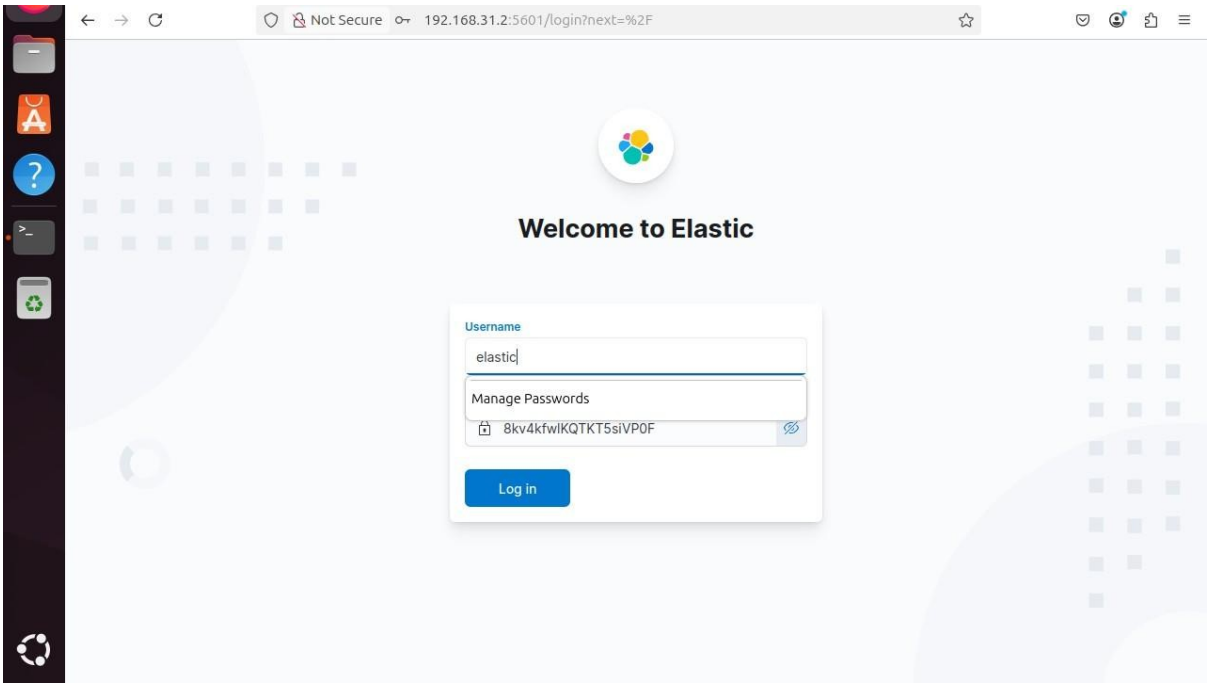
Change the path to /opt/zeek/share/zeek/site/local.zeek/*.log

nano /etc/filebeat/modules.d/zeek.yml (*change the path*)

```
GNU nano 7.2                    /etc/filebeat/modules.d/zeek.yml
# Module: zeek
# Docs: https://www.elastic.co/guide/en/beats/filebeat/7.x/filebeat-module-zeek.html
- module: zeek
  capture_loss:
    enabled: true
    var.paths: ["/opt/zeek/logs/current/capture_loss.log"]
  connection:
    enabled: true
    var.paths: ["/opt/zeek/logs/current/conn.log"]
  dce_rpc:
    enabled: true
    var.paths: ["/opt/zeek/logs/current/dce_rpc.log"]
  dhcp:
    enabled: true
    var.paths: ["/opt/zeek/logs/current/dhcp.log"]
  dnp3:
    enabled: true
    var.paths: ["/opt/zeek/logs/current/dnp3.log"]
  dns:
    enabled: true
    var.paths: ["/opt/zeek/logs/current/dns.log"]
  dpd:
    enabled: true
    var.paths: ["/opt/zeek/logs/current/dpd.log"]
  files:
    enabled: true
    var.paths: ["/opt/zeek/logs/current/files.log"]
  ftp:
                                    [ Read 123 lines ]
```

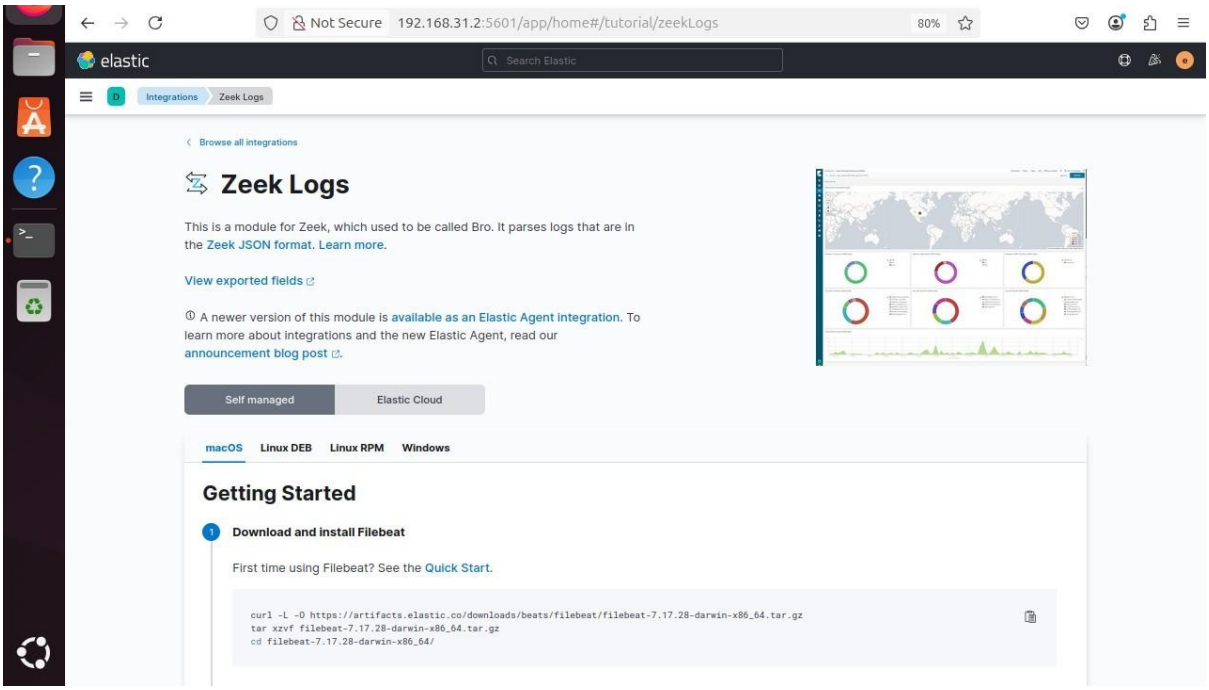
Follow the as per the above image

Search http://ubantuip:5601 and login with credentials

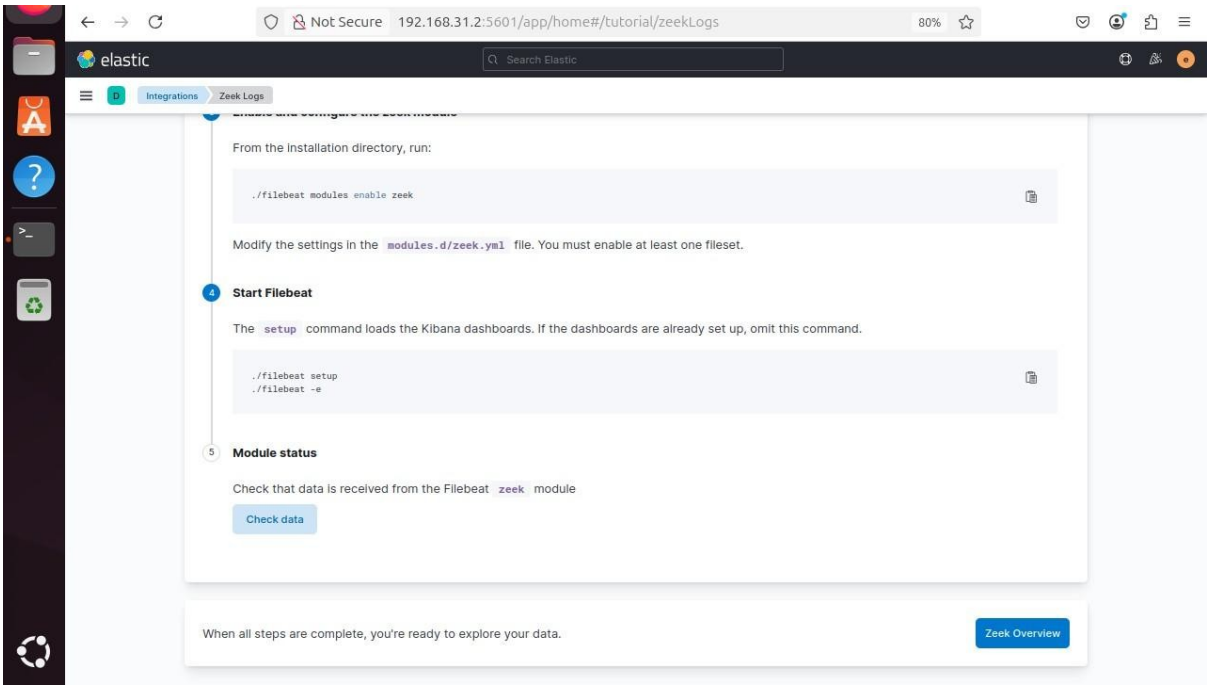


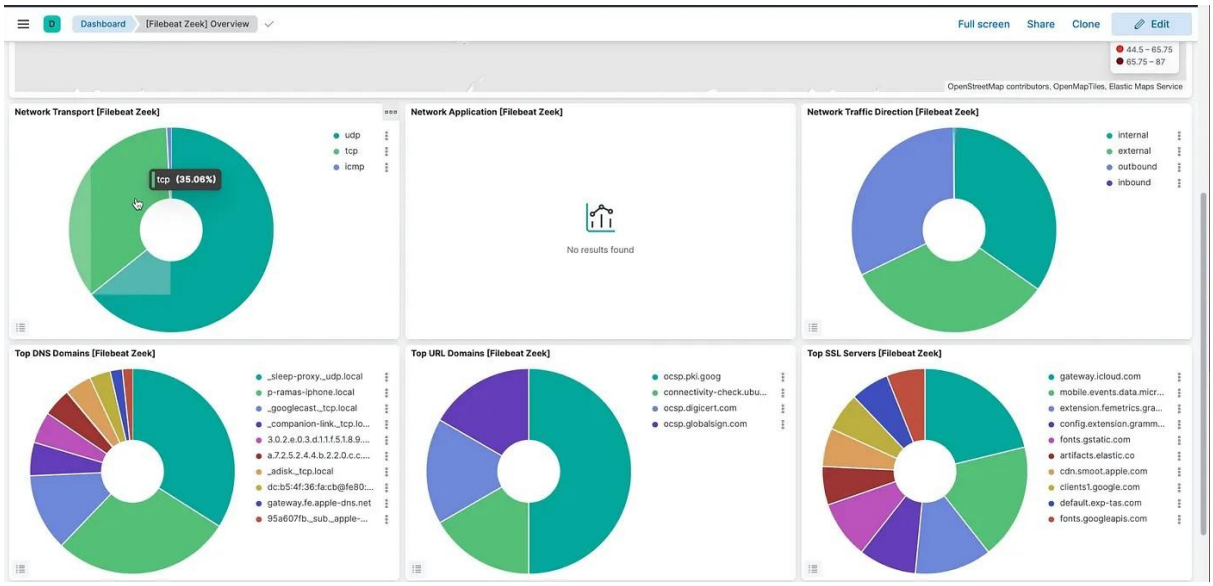
Go in integration section and search for zeeklogs

Copy the commands and paste them on the terminal



1. Install Filebeat: Set up Filebeat on the server where Zeek is running to ship logs to Elasticsearch.
2. Configure Filebeat: Edit the Filebeat configuration file to specify the path to the Zeek logs and set the output to Elasticsearch.
3. Start Filebeat: Run Filebeat to begin sending logs to Elasticsearch.
4. Access Kibana: Use Kibana to visualize and analyze the Zeek logs.



Follow the steps as per above

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